DTC P2002:00 [SKYACTIV-D 1.5]

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Note
- To determine the malfunctioning part, proceed with the diagnostics from “Function Inspection Using M-MDS”.

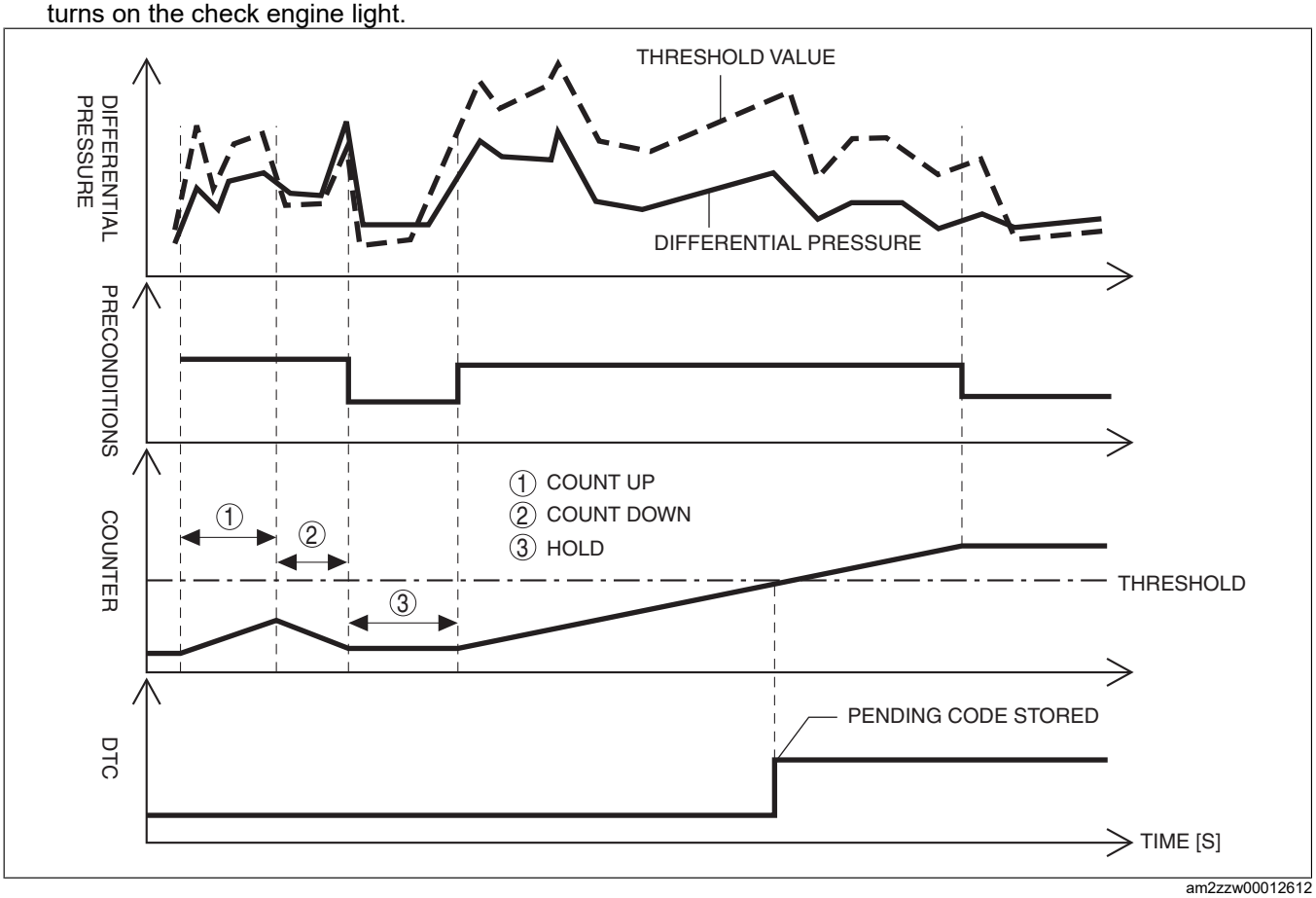
Details On DTCs

DESCRIPTION	Diesel particulate filter function decreased	
DETECTION CONDITION	Determination conditions	• The difference in the pressure before and after passing the diesel particulate filter of less than the specified value is detected for **2s**.
	Preconditions	• All of the following conditions are met for **2s**: — APP: **0.5% or less** — Fuel injection amount control: during fuel cut — Amount of exhaust gas: **1.1 m^3/min or more** — Selector lever drive position and clutch engagement (MTX) — Selector lever drive range (ATX) — 300 s or more elapsed since DPF regeneration control stopped — Engine coolant temperature after engine start: **40 °C {104 °F} or more** — Intake air temperature: **–10—110 °C {14—230 °F}** — Barometric pressure: **70—110 kPa {0.72—1.12 kgf/cm^2, 11.0—15.9 psi}** — Amount of change in engine speed: **40 rpm or less** for **1 s** — Amount of change in exhaust gas: **1.8 m^3/min or more** for **1 s** — Amount of change in pressure applied to diesel particulate filter: **0.62 kPa {0.0063 kgf/cm^2, 0.090 psi} or less** for **1 s** — The traveled distance after DPF regeneration exceeds **21 km {14 mile}** and **10 s** have elapsed with DPF temperature of **350 °C {662 °F} or more**, or traveled distance **exceeds 500 km {311 mile}**. — Traveled distance after DPF regeneration **exceeds 1000 km {621.4 mile}** or accumulated PM amount **exceeds 3.4 g/l {0.22 lb/ft^3}**, or it is **less than 1 g/l {0.062 lb/ft^3}**
	Drive cycle	• **2**
	Self test type	• CMDTC self test
	Sensor used	• Exhaust gas pressure sensor No.1 • Exhaust gas pressure sensor No.2 etc.
FAIL-SAFE FUNCTION	• Not applicable	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Check engine light is illuminated	
POSSIBLE CAUSE	• Exhaust gas leakage from exhaust system • Pipe between exhaust gas pressure sensor No.2 and catalytic converter (diesel particulate filter) restriction and/or damaged or improper routing • Exhaust gas pressure sensor No.2 malfunction • Exhaust gas pressure sensor No.1 malfunction • Catalytic converter (diesel particulate filter) or diesel particulate filter malfunction (deformation, damage) • PCM malfunction	

System Wiring Diagram
- Not applicable

Function Explanation (DTC Detection Outline)
- The PCM performs diagnosis of the DPF soot collection performance by comparing the exhaust gas pressure before and after the DPF.
- The PCM uses exhaust gas pressure sensor No.2 to monitor the pressure differences (actual pressure differences) before and after the DPF.
- When the monitoring conditions are met, the PCM compares the DPF before and after pressure differences based on the calculation of the actual pressure difference and the exhaust gas flow volume. If the actual pressure difference at this time increases above or decreases below the specified value relative to the calculated pressure difference, the PCM determines that the DPF soot collection performance has lowered.

- The PCM performs diagnosis when each of the preconditions during drive cycle is met. If a malfunction is detected, the malfunction detection counter begins counting up. The value of the malfunction detection counter is maintained while the preconditions are not being met. If the cumulative value of the malfunction detection counter exceeds the threshold during the first drive cycle, a malfunction is determined and a pending code is stored.
- If the PCM determines that the malfunction recurs from the next drive cycle and thereafter, it stores a DTC and turns on the check engine light.



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Repeatability Verification Procedure

Caution
- **When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.**
- **While performing this step, always operate the vehicle in a safe and lawful manner.**

Note
- If the DPF (catalyst converter) is replaced as a result of the diagnosis, perform the procedure from Step 3.

1. Perform compulsory DPF regeneration. (See COMPULSORY DIESEL PARTICULATE FILTER REGENERATION [SKYACTIV-D 1.5].)
2. Stop the engine and leave for 1 hour or more.
3. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].)
4. After starting the engine, drive the vehicle for approx. 30 km {19 mile} and warm up the engine.
5. Release the accelerator pedal and decelerate the vehicle for 5 s or more, after rapidly accelerate the vehicle to 120 km/h {74.6 mph}.
6. Repeat Step 5 operation above 2 times.

PID Item/Simulation Item Used In Diagnosis
- Not applicable

Function Inspection Using M-MDS

STEP	INSPECTION		ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Is the DTC P2002:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-D 1.5].)
3	**PURPOSE: RECORD FREEZE FRAME DATA/ SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (misfire related) on the repair order.	—	Go to the next step.
4	**PURPOSE: VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].) Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1
  - — Perform inspection of exhaust gas leakage from exhaust system.
- Step 2
  - — Perform inspection of exhaust gas temperature sensor No.2 related pipe connection.
- Step 3—4
  - — Perform inspection of each separate part.
- Step 5—6
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

Function Inspection Using M-MDS

STEP	INSPECTION		ACTION
1	**PURPOSE: INSPECT EXHAUST SYSTEM FOR LEAKAGE** • Visually inspect for exhaust gas leakage from the exhaust system. • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 5.
		No	Go to the next step.
2	**PURPOSE: INSPECT EXHAUST GAS PRESSURE SENSOR NO.2 RELATED PIPE** • Visually inspect the exhaust gas pressure sensor No.2 related pipe for restriction and damaged. (See EXHAUST GAS PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 5.
		No	Go to the next step.
3	**PURPOSE: INSPECT EXHAUST GAS PRESSURE SENSOR NO.2** • Reconnect all disconnected connectors. • Inspect the exhaust gas pressure sensor No.2. (See EXHAUST GAS PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the exhaust gas pressure sensor No.2, then go to Step 5. (See EXHAUST GAS PRESSURE SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**PURPOSE: INSPECT EXHAUST GAS PRESSURE SENSOR NO.1** • Reconnect all disconnected connectors. • Inspect the exhaust gas pressure sensor No.1. (See EXHAUST GAS PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the exhaust gas pressure sensor No.1, then go to the next step. (See EXHAUST GAS PRESSURE SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Catalytic converter (diesel particulate filter) or diesel particulate filter can be considered the cause. • Replace the catalytic catalytic converter (diesel particulate filter), then go to the next step. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5	**PURPOSE: PERFORM DTC INSPECTION AND VERIFY IF MALFUNCTIONING PART IS PCM** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
6	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.